

Amitabh Shukla

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 [AMITABHSHUKLA](#)

TECHNICAL SKILLS

Programming Language: Python

ML: Supervised Learning, Unsupervised Learning, Model Selection, Feature Engineering, pandas, NumPy

Deep Learning: PyTorch, ViT, Transformers, Vision Language Models, CLIP, SigLIP, PaliGemma, MemGPT

LLMs and Computer Vision: LLM Fine-Tuning (LoRA, QLoRA, PEFT), KV Cache, Paged KV Cache, Grouped Query Attention, Flash Attention, MQA, RoPE, Chain-of-Thought, Mixture of Experts, OpenCV, Faster R-CNN

Backend and Databases: FastAPI, REST APIs, PostgreSQL, SQLite, SQLAlchemy ORM

Cloud and DevOps: AWS SageMaker, Bedrock, Lambda, S3, GitHub Actions, Git, GitHub

EXPERIENCE

AI/ML Engineer II, Kreeda Labs, Pune

03/2026 – Present

Multimodal Video Intelligence System | **Multiversity** | LLM • CV • RAG

- Built a **multimodal video-to-action intelligence system** that learns task workflows from raw videos by aligning fine-grained visual actions with audio semantics, converting unstructured content into structured, executable knowledge.
- Engineered a **signal-preserving perception pipeline** (SSIM, pixel-difference, embedding filtering) to isolate action-critical frames and eliminate noise, ensuring high-fidelity step-level understanding, achieving **98% action-level accuracy with 95.44% cost reduction**.
- Developed a **multimodal retrieval layer** (hybrid BM25 + dense embeddings, multi-vector image representations, re-ranking) to reconstruct cross-segment context and dependencies.
- Converted outputs into **Video Description Schemas (VDS)** to drive **autonomous AI agents** executing complex software workflows without human intervention.

ML Engineer, Kynea Solutions LLP, Hyderabad

09/2025 – 02/2026

Intelligent Ticket Routing and LLM Support System | **Lenovo India** | NLP • LLM • AWS • MLOps

- Designed and deployed an enterprise NLP platform with multi-stage classification (10 categories, 14 subcategories) using Python, PyTorch, Transformers, and Scikit-learn, achieving 90%+ accuracy and serving real-time predictions via Amazon SageMaker.
- Built a deterministic agent-ranking system leveraging 20+ expertise, workload, SLA, and performance signals to fairly route tickets across 10k–30k agents, replacing legacy random assignment at production scale.
- Fine-tuned LLaMA 3.1 8B with QLoRA (PEFT) on 50k support conversations and integrated a FAISS-based RAG stack (LangChain, LangGraph, AWS Bedrock), deployed via Terraform and CI/CD, reducing routing latency by 32% with automated model versioning.

Build OCR and Object Detection Pipeline for Engineering Diagrams | **Motherson Group** | CV • OCR • Azure

- Designed an end-to-end OCR and object detection system for ultra high-resolution engineering diagrams, achieving 96% OCR accuracy and 94% mAP using Azure OCR and Faster R-CNN with custom post-processing for layout reconstruction, bounding-box correction, and automated BOM generation.

PROJECTS

VidyaAI 

- Built a multimodal learning assistant leveraging GPT-4o, Whisper, and Retrieval-Augmented Generation (RAG) to deliver fast, context-aware answers across documents and YouTube content.
- Designed a unified retrieval pipeline enabling multi-document querying, video understanding, and retrieval-driven conversational AI for enhanced knowledge access.
- Improved learning efficiency for students and professionals by structuring unorganized content into accurate, actionable responses.

EDUCATION

- **Oriental Institute of Science and Technology**, Bhopal, M.P.
B.Tech in Computer Science and Engineering (AIML)

2021 – 2025